

H950 Hardened S46500 Stainless Steel

H950 hardened S46500 stainless steel is S46500 stainless steel in the hardened (H) condition. It has the second highest strength compared to the other variants of S46500 stainless steel.

The graph bars on the material properties cards below compare H950 hardened S46500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Brinell Hardness

500

Elastic (Young's, Tensile) Modulus

190 GPa

28 x 10⁶ psi

Elongation at Break

9.1 %

Fatigue Strength

890 MPa	130 x 10 ³ psi
Poisson's Ratio	
0.28	
Reduction in Area	
39 %	
Rockwell C Hardness	
54	
Shear Modulus	
75 GPa	11 x 10 ⁶ psi
Shear Strength	
1120 MPa	160 x 10 ³ psi
Tensile Strength: Ultimate (UTS)	
1890 MPa	270 x 10 ³ psi
Tensile Strength: Yield (Proof)	
1700 MPa	250 x 10 ³ psi

Thermal Properties

Latent Heat of Fusion	
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280 J/g

Maximum Temperature: Corrosion

680 °C

1260 °F

Maximum Temperature: Mechanical

780 °C

1430 °F

Melting Completion (Liquidus)

1450 °C

2650 °F

Melting Onset (Solidus)

1410 °C

2570 °F

Specific Heat Capacity

470 J/kg-K

0.11 BTU/lb-°F

Thermal Expansion

11 µm/m-K

Otherwise Unclassified Properties

Base Metal Price

15 % relative

Density

7.9 g/cm ³	490 lb/ft ³
Embodied Carbon	
3.6 kg CO ₂ /kg material	
Embodied Energy	
51 MJ/kg	22 x 10 ³ BTU/lb
Embodied Water	
120 L/kg	14 gal/lb

Common Calculations

PREN (Pitting Resistance)	
15	
Resilience: Ultimate (Unit Rupture Work)	
170 MJ/m ³	
Stiffness to Weight: Axial	
14 points	
Stiffness to Weight: Bending	
25 points	
Strength to Weight: Axial	

67 points

Strength to Weight: Bending

43 points

Thermal Shock Resistance

65 points

Alloy Composition

Iron (Fe)	72.6 to 76.1
Chromium (Cr)	11 to 12.5
Nickel (Ni)	10.7 to 11.3
Titanium (Ti)	1.5 to 1.8
Molybdenum (Mo)	0.75 to 1.3
Manganese (Mn)	0 to 0.25
Silicon (Si)	0 to 0.25
Carbon (C)	0 to 0.020
Phosphorus (P)	0 to 0.015
Sulfur (S)	0 to 0.010
Nitrogen (N)	0 to 0.010

All values are % weight. Ranges represent what is permitted under applicable standards.

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM A564: Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

Corrosion of Stainless Steels, A. John Sedriks, 1996

ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

Advances in Stainless Steels, Baldev Raj et al. (editors), 2010

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